



Topic	Task	Function / Syntax	Small Definition
Import	Import Pandas	<code>import pandas as pd</code>	Import the Pandas library
Data Structures	Series	<code>pd.Series()</code>	Create a one-dimensional labeled array
	DataFrame	<code>pd.DataFrame()</code>	Create a two-dimensional table
Reading Data	Read CSV	<code>pd.read_csv()</code>	Read data from a CSV file
	Read Excel	<code>pd.read_excel()</code>	Read data from an Excel file
	Read JSON	<code>pd.read_json()</code>	Read data from a JSON file
	Read SQL	<code>pd.read_sql()</code>	Read data from a SQL database
Writing Data	Write CSV	<code>df.to_csv()</code>	Save DataFrame to a CSV file
	Write Excel	<code>df.to_excel()</code>	Save DataFrame to an Excel file
	Write JSON	<code>df.to_json()</code>	Save DataFrame to a JSON file
Data Inspection	First Rows	<code>df.head()</code>	Display the first rows
	Last Rows	<code>df.tail()</code>	Display the last rows
	Shape	<code>df.shape</code>	Returns number of rows and columns
	Size	<code>df.size</code>	Total number of elements
	Dimensions	<code>df.ndim</code>	Number of dimensions
	Column Names	<code>df.columns</code>	Returns column names
	Index	<code>df.index</code>	Returns row labels
	Data Types	<code>df.dtypes</code>	Returns data types of columns
	Information	<code>df.info()</code>	Displays DataFrame information
Selecting Data	Statistics	<code>df.describe()</code>	Summary statistics of numeric columns
	Single Column	<code>df["Name"]</code>	Select one column

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Filtering Data	Multiple Columns	<code>df[["Name", "Age"]]</code>	Select multiple columns
	Row by Label	<code>df.loc[]</code>	Access rows using labels
	Row by Position	<code>df.iloc[]</code>	Access rows using index positions
	Filter Rows	<code>df[df["Age"]>25]</code>	Filter rows using conditions
	Multiple Conditions	<code>&, \ </code>	Combine multiple conditions
Adding & Modifying	Query	<code>df.query()</code>	Filter using query expressions
	Add Column	<code>df["City"]="Hyderabad"</code>	Create a new column
	Rename Column	<code>df.rename()</code>	Rename columns
	Change Data Type	<code>df.astype()</code>	Convert column data type
Removing Data	Drop Column	<code>df.drop(columns=[])</code>	Remove one or more columns
	Drop Row	<code>df.drop(index=[])</code>	Remove one or more rows
	Remove Duplicates	<code>df.drop_duplicates()</code>	Delete duplicate rows
Missing Values	Find Missing Values	<code>df.isnull()</code>	Detect missing values
	Count Missing Values	<code>df.isnull().sum()</code>	Count missing values
	Fill Missing Values	<code>df.fillna()</code>	Replace missing values
	Remove Missing Values	<code>df.dropna()</code>	Remove missing values
Sorting	Sort Values	<code>df.sort_values()</code>	Sort rows by column values
	Sort Index	<code>df.sort_index()</code>	Sort by row index
Aggregation	Sum	<code>df.sum()</code>	Calculate sum
	Mean	<code>df.mean()</code>	Calculate average
	Maximum	<code>df.max()</code>	Largest value

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	Minimum	<code>df.min()</code>	Smallest value
	Count	<code>df.count()</code>	Count non-null values
	Value Counts	<code>df.value_counts()</code>	Count unique values
Grouping	Group By	<code>df.groupby()</code>	Group rows based on column values
	Aggregate	<code>df.agg()</code>	Apply multiple aggregate functions
Combining Data	Merge	<code>pd.merge()</code>	Combine DataFrames using common columns
	Concatenate	<code>pd.concat()</code>	Combine DataFrames vertically or horizontally
	Join	<code>df.join()</code>	Join DataFrames using index
String Operations	Lowercase	<code>df["col"].str.lower()</code>	Convert text to lowercase
	Uppercase	<code>df["col"].str.upper()</code>	Convert text to uppercase
	Replace Text	<code>df["col"].str.replace()</code>	Replace text values
	Contains	<code>df["col"].str.contains()</code>	Check if text contains a value
Date & Time	Convert Date	<code>pd.to_datetime()</code>	Convert values to datetime
	Current Date	<code>pd.Timestamp.now()</code>	Get current date and time
Apply Functions	Apply Function	<code>df.apply()</code>	Apply function to rows or columns
	Map Values	<code>df["col"].map()</code>	Map values using dictionary or function
	Apply Element	<code>df.applymap()</code>	Apply function to every element
Index Operations	Set Index	<code>df.set_index()</code>	Set a column as index
	Reset Index	<code>df.reset_index()</code>	Reset index to default
Exporting Data	Save CSV	<code>df.to_csv()</code>	Export DataFrame to CSV
	Save Excel	<code>df.to_excel()</code>	Export DataFrame to Excel
	Save JSON	<code>df.to_json()</code>	Export DataFrame to

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JSON			

Function	Purpose
<code>pd.read_csv()</code>	Read CSV file
<code>pd.read_excel()</code>	Read Excel file
<code>pd.DataFrame()</code>	Create DataFrame
<code>df.head()</code>	First 5 rows
<code>df.tail()</code>	Last 5 rows
<code>df.info()</code>	Dataset information
<code>df.describe()</code>	Summary statistics
<code>df.shape</code>	Rows and columns
<code>df.columns</code>	Column names
<code>df.dtypes</code>	Data types
<code>df.loc[]</code>	Label-based selection
<code>df.iloc[]</code>	Position-based selection
<code>df.drop()</code>	Remove rows or columns
<code>df.rename()</code>	Rename columns
<code>df.sort_values()</code>	Sort data
<code>df.groupby()</code>	Group data
<code>df.merge()</code>	Merge DataFrames
<code>pd.concat()</code>	Concatenate DataFrames
<code>df.isnull()</code>	Find missing values
<code>df.fillna()</code>	Fill missing values
<code>df.dropna()</code>	Remove missing values
<code>df.value_counts()</code>	Count unique values
<code>df.sum()</code>	Sum values
<code>df.mean()</code>	Average values
<code>df.max()</code>	Maximum value
<code>df.min()</code>	Minimum value
<code>df.apply()</code>	Apply custom function

Function	Purpose
<code>df.to_csv()</code>	Save CSV file
<code>df.to_excel()</code>	Save Excel file

most frequently asked Pandas interview question

Concept	Interview Question	Simple Interview Answer
Pandas	What is Pandas?	Pandas is a Python library used for data analysis, data cleaning, and data manipulation. It mainly works with Series and DataFrames.
Series vs DataFrame	Difference between Series and DataFrame?	A Series is a one-dimensional data structure. A DataFrame is a two-dimensional table with rows and columns.
DataFrame	What is a DataFrame?	A DataFrame is the main Pandas data structure that stores data in rows and columns, similar to an Excel sheet or SQL table.
Reading Data	How do you read a CSV file in Pandas?	Use <code>pd.read_csv()</code> to read a CSV file into a DataFrame.
Reading Excel	How do you read an Excel file?	Use <code>pd.read_excel()</code> to read an Excel file.
head() vs tail()	What are <code>head()</code> and <code>tail()</code> ?	<code>head()</code> displays the first 5 rows, while <code>tail()</code> displays the last 5 rows by default.
info()	What is <code>info()</code> used for?	It displays column names, data types, non-null values, and memory usage.
describe()	What does <code>describe()</code> do?	It returns summary statistics such as count, mean, minimum, maximum, and standard deviation.
shape	What does <code>shape</code> return?	It returns the number of rows and columns in a DataFrame.
columns	What does <code>columns</code> return?	It returns the names of all columns in the DataFrame.
dtypes	What is <code>dtypes</code> ?	It shows the data type of each column.
loc vs iloc	Difference between <code>loc[]</code> and <code>iloc[]</code> ?	<code>loc[]</code> selects data using row or column labels. <code>iloc[]</code> selects data using index positions.
Filtering	How do you filter rows in Pandas?	Use conditions like <code>df[df["Age"] > 25]</code> to filter rows.
Sorting	How do you sort a DataFrame?	Use <code>df.sort_values()</code> to sort rows based on column values.
Missing Values	How do you find missing values?	Use <code>df.isnull()</code> or <code>df.isnull().sum()</code> .

Concept	Interview Question	Simple Interview Answer
fillna()	What is <code>fillna()</code> ?	It replaces missing values with a specified value.
dropna()	What is <code>dropna()</code> ?	It removes rows or columns that contain missing values.
drop()	What is <code>drop()</code> ?	It removes specified rows or columns from a DataFrame.
rename()	What is <code>rename()</code> ?	It changes the names of columns or indexes.
groupby()	What is <code>groupby()</code> ?	It groups rows with the same values to perform calculations like sum, mean, and count.
merge() vs concat()	Difference between <code>merge()</code> and <code>concat()</code> ?	<code>merge()</code> joins DataFrames using common columns. <code>concat()</code> combines DataFrames vertically or horizontally.
apply()	What is <code>apply()</code> used for?	It applies a custom function to rows or columns of a DataFrame.
value_counts()	What is <code>value_counts()</code> ?	It counts how many times each unique value appears in a column.
set_index()	What is <code>set_index()</code> ?	It sets a column as the row index.
reset_index()	What is <code>reset_index()</code> ?	It restores the default numeric index.
Export Data	How do you save a DataFrame?	Use <code>df.to_csv()</code> or <code>df.to_excel()</code> to save the DataFrame.
Pandas Advantages	Why is Pandas popular?	Pandas is fast, easy to use, supports large datasets, and provides powerful tools for data analysis and cleaning.